

Project Title:-

Suspicious Encoded PowerShell Execution.

Executive Summary:-

Investigated suspicious PowerShell execution launched through cmd.exe using encoded commands and evasion flags. Analysis identified the use of -nop, -w hidden and -enc, commonly associated with malicious PowerShell activity.

Environment:-

- Tool Used: Sysmon
- Data Source: Process Log
- Platform: Splunk

Alert Overview:-

- The alert was triggered due to suspicious PowerShell execution launched from cmd.exe using encoded commands and evasion flags (-nop, -w hidden, -enc).
- The Powershell flags -nop, -w hidden indicated attempts to evade detection and operate stealthily.

Investigation Process:-

- Reviewed Sysmon Process Creation logs
- Identified powershell execution
- Examined parent-child relationship
- Analyzed command line arguments
- Investigated encoded command
- Reviewed PowerShell execution flags

Response Actions:-

- Preserved evidence
- Terminated the suspicious PowerShell process
- Reviewed host for additional suspicious activity
- Monitored for further PowerShell execution
- Recommended further endpoint monitoring

Key Findings:-

- Powershell flags (-nop, -w hidden, -enc)
- CommandLine (powershell.exe -nop -w hidden -enc
SUVYIChOZctT2JqZWN0IE5IdC5XZWJDbGIbnQpLkRvd25sb2Fk....)

Indicators of Compromise (IOCs):-

Process: powershell.exe

Parent Process: cmd.exe

PowerShell Flags: -nop, -w hidden, -enc

User: SOC-Lab

Host: DESKTOP-6A40QRO

Timeline:-

16:46:28 – powershell executed

16:46:28 – Alert reviewed in splunk

16:48:00 – Parent-child analysis performed

16:48:00 – Host reviewed for additional persistence

16:52:00 – Investigation closed

Detection Logic:-

Suspicious powershell.exe execution:

Combined with PowerShell evasion flags:

-nop

-w hidden

-enc

MITRE ATT&CK Mapping:-

- T1059.001 – PowerShell
- T1027 – Obfuscated/Encoded Files and Information

Lessons Learned:-

- Always check the powershell activity with the context although it is a legitimate feature use by administrators but abuse by the attackers mostly with combination of -enc.

Skills Demonstrated:-

- Log Analysis
- Splunk
- Incident Response
- Threat Hunting
- Documentation

Severity:-

Medium

Reason is, PowerShell was executed using encoded commands and evasion flags, but no additional malicious activity was identified during the investigation.

Evidence Collected:-

1. Encoded PowerShell Execution

6/7/26
4:46:27786 PM

```
<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Sysmon' Guid='{5770385f-c22a-43e0-bf4c-06f5698ffbd9}' /><EventID>1</EventID><Version>5</Version><Level>4</Level><Task>1</Task><Opcode>0</Opcode><Keywords>0x8000000000000000</Keywords><TimeCreated SystemTime='2026-06-07T15:46:27.7864606 Z' /><EventRecordID>50227</EventRecordID><Correlation><Execution ProcessID='3008' ThreadID='4576' /><Channel>Microsoft-Windows-Sysmon/Operational</Channel><Computer>DESKTOP-6A40QRO</Computer><Security UserID='S-1-5-18' /></System><EventData><Data Name='RuleName' /></Data><Data Name='UtcTime' />2026-06-07 15:46:27.171</Data><Data Name='ProcessGUID'>{elbdfed9-9253-6a25-9001-000000004500}</Data><Data Name='ProcessId'>4968</Data><Data Name='Image'>C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe</Data><Data Name='FileVersion'>10.0.19041.3996 (WinBuild.160101.0800)</Data><Data Name='Description'>Windows PowerShell</Data><Data Name='Product'>Microsoft Windows Operating System</Data><Data Name='Company'>Microsoft Corporation</Data><Data Name='OriginalFileName'>PowerShell.EXE</Data><Data Name='CommandLine'>powershell.exe -nop -w hidden -enc SUVYIChOZXCtT2JqZWNOIE5ldCSXZWJDsGllbnQpLkRvdWZ5b2FkuU3RyaW5kcDdhIRwOI8vMTI3LjAuMC4xL3Ric3</Data><Data Name='CurrentDirectory'>C:\Windows\system32</Data><Data Name='User'>DESKTOP-6A40QRO\SOC-Lab</Data><Data Name='LogonGUID'>{elbdfed9-9154-6a25-630e-060000000000}</Data><Data Name='LogonId'>0x60e63c</Data><Data Name='TerminalSessionId'>1</Data><Data Name='IntegrityLevel'>High</Data><Data Name='Hashes'>SHA256=9785001B80CF755ED0B8AF294A373CB087B2498660F72AE76C4D53FC9C217C7A3</Data><Data Name='ParentProcessGUID'>{elbdfed9-9154-6a25-e700-000000004500}</Data><Data Name='ParentProcessId'>7872</Data><Data Name='ParentImage'>C:\Windows\System32\cmd.exe</Data><Data Name='ParentCommandLine'>'C:\Windows\system32\cmd.exe'</Data><Data Name='ParentUser'>DESKTOP-6A40QRO\SOC-Lab</Data></EventData></Event>
```

Event Actions ▾

Type	Field	Value	Actions
Selected	☒ host ▾	DESKTOP-6A40QRO	
	☒ source ▾	WinEventLog:Microsoft-Windows-Sysmon/Operational	
	☒ sourcetype ▾	XmlWinEventLog:Microsoft-Windows-Sysmon/Operational	
	☐ CommandLine ▾	powershell.exe -nop -w hidden -enc SUVYIChOZXCtT2JqZWNOIE5ldCSXZWJDsGllbnQpLkRvdWZ5b2FkuU3RyaW5kcDdhIRwOI8vMTI3LjAuMC4xL3Ric3QnKQ==	
Event	☐ Image ▾	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	
	☐ ParentImage ▾	C:\Windows\System32\cmd.exe	
	☐ User ▾	DESKTOP-6A40QRO\SOC-Lab	
	☐ Time ▾	2026-06-07T16:46:27786+0100	
Default	☐ Index ▾	main	
	☐ lincount ▾	1	
	☐ punct ▾	<=> , /><-_-'-_-[]->/></></></></>	
	☐ splunk_server ▾	LAPTOP-CHVDYIONG	

2. Parent Child Relationship

☐ Image ▾	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
☐ ParentImage ▾	C:\Windows\System32\cmd.exe

3. PowerShell Evasion Flags

```
6/7/26 <Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Sysmon' Guid='{5770385f-c22a-43e0-bf4c-06f5698fbfd9}'><EventID>1</EventID><Version>5</Version><Level>4</Level><Task>1</Task><Opcode>0</Opcode><Keywords>0x8000000000000000</Keywords><TimeCreated SystemTime='2026-06-07T11:45:26.784606Z' /><EventRecordID>50227</EventRecordID><Correlation><Execution ProcessID='3008' ThreadID='4576' /><Channel>Microsoft-Windows-Sysmon/Operational</Channel><Computer>DESKTOP-06A40RQ</Computer><Security UserID='S-1-5-18' /></System><EventData><Data Name='RuleName'></Data><Data Name='UtcTime'>2026-06-07 15:46:27.171</Data><Data Name='ProcessGuid'>[e1bf9d9-9253-6a25-90d1-000000004500]</Data><Data Name='ProcessID'>4968</Data><Data Name='Image'><C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe</Data><Data Name='FileVersion'>10.0.19041.3996 (WinBuild.160101.0800)</Data><Data Name='Description'>Windows PowerShell</Data><Data Name='Product'>Microsoft's Windows Operating System</Data><Data Name='Company'>Microsoft Corporation</Data><Data Name='OriginalFileName'>PowerShell.EXE</Data><Data Name='CommandLine'>powershell.exe -nop -w hidden -enc SUIVCH0ZQZt72JqZWNoFE5d0c5XZjWdG1b1n0pKkrV25ZfKUR3v5mKd0dH0i0wMTI31j4uCAmL3Rl3c0NkQ==</Data><Data Name='CurrentDirectory'>C:\Windows\system32</Data><Data Name='User'>DESKTOP-06A40RQ\SOC-Lab</Data><Data Name='LogonGUID'>[e1bf9d9-9154-6a25-630e-000000000000]</Data><Data Name='LogonID'>0xb06b3</Data><Data Name='TerminalSessionID'>1</Data><Data Name='IntegrityLevel'>High</Data><Data Name='Hashes'>SHA256=9785001B0DC7F55EDB8A729A437C08B78498660774267F64D539321C71A3</Data><Data Name='ParentProcessGUID'>[e1bf9d9-9165-6a25-e080-000000004500]</Data><Data Name='ParentProcessID'>7872</Data><Data Name='ParentImage'><C:\Windows\System32\cmd.exe</Data><Data Name='ParentCommandLine'><C:\Windows\system32\cmd.exe /></Data><Data Name='ParentUser'>DESKTOP-06A40RQ\SOC-Lab</Data></EventData></Event>
```

Analyst Verdict:-

Suspicious PowerShell execution was confirmed.

Evidence showed PowerShell launched using encoded commands and evasion n flags (-nop, -w hidden, -enc). While the activity matched common tradecraft, no additional malicious actions were observed during the investigation. The event was classified as suspicious and required further monitoring.